

Notes for your Keras3/Tensorflow/pytorch environnement

Hands-on 2 Classification



— Montreal, June 15–19 —

DLMI2026

Preliminary remarks

- All four hands-on can be done on a computing cloud such as that of *CalculQuebec*, *Google Colab*, ... or on your own computer
- **For beginners** : we recommend you to start with *CalculQuebec*
- For advanced users with CPU/GPU access
 - You can run on your specific hardware
 - Tips for installing the required environnement are [here](#).

First, setup your *CalculQuebec* Account

(cf. DLMI 2026 hands on setup instructions)

1. *Create an account on CalculQuebec Cloud and log-in*
<https://jupyter.dmlti.calculquebec.cloud/hub/login>
2. Create a compute node with
 1. Memory at 4096MB
 2. GPU 1x1G.5GB
3. Start the jupyter-lab server and wait for it **Steps covered here**
4. Download the hands-on material into your jupyter-lab
5. Open a notebook
6. Cells philosophy and execution

Jupyter Lab interface (can change)

The screenshot shows the Jupyter Lab web interface. On the left is a file browser sidebar with a table of files and folders. Two orange arrows point to the 'upload' button (a square with an upward arrow) and the file list area. A red text box at the bottom left says "Drag and drop files here (or via the upload button)". The main area is titled "Launcher" and contains several sections: "Notebook" with icons for Python 3.11, LibreQDA, Mate Desktop (VNC), MLflow, OpenRefine, Python bundle, RStudio, Tensorboard, and VS Code; "Console" with Python 3.11 and Python bundle; "Other" with icons for Terminal, Job statistics, Text File, Markdown File, Python File, and Show Contextual Help; and "Documentation" with General and Jupyter links. A green arrow points to the "Launcher" tab at the top, and another green arrow points to the "Terminal" icon in the "Other" section. A small notification box in the bottom right corner says "Would you like Jupyter news?".

File browser sidebar:

Name	Modified
projects	last mo.
scratch	last mo.

Launcher tab

Terminal

Drag and drop files here (or via the upload button)

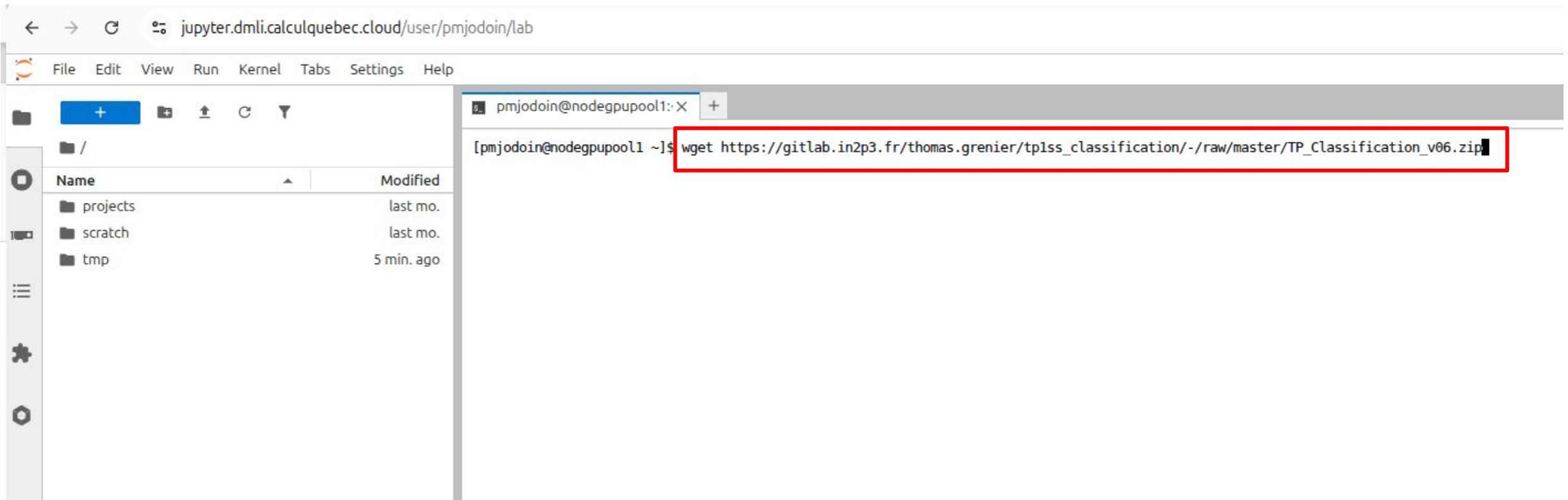
Download Hands-on materials

(same as for hands on 1)

a- Open a **Terminal** (c.f. previous page)

b- download from the terminal with the following command :

```
wget https://gitlab.in2p3.fr/thomas.grenier/tp1ss_classification/-/raw/master/TP_Classification_v06.zip
```



Download Hands-on materials

a- Open a **Terminal**

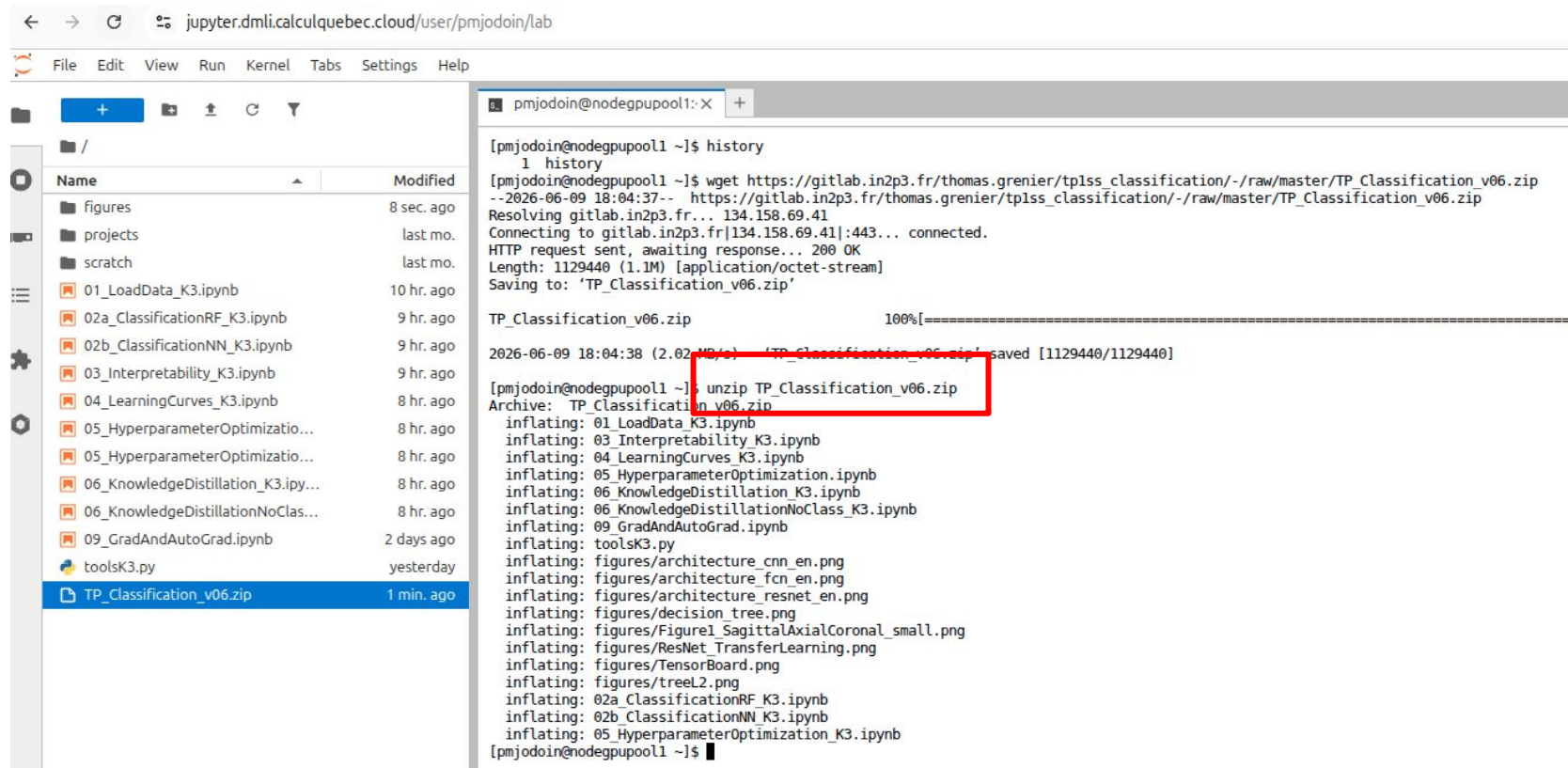
b- download from the terminal with :

```
wget https://gitlab.in2p3.fr/thomas.grenier/tp1ss_classification/-/raw/master/TP_Classification_v06.zip
```

c- unzip the file (in a Terminal)

```
unzip TP_Classification_v06.zip
```

unzipped files



```
[pmjodoin@nodegpupool1 ~]$ history
1 history
[pmjodoin@nodegpupool1 ~]$ wget https://gitlab.in2p3.fr/thomas.grenier/tp1ss_classification/-/raw/master/TP_Classification_v06.zip
--2026-06-09 18:04:37-- https://gitlab.in2p3.fr/thomas.grenier/tp1ss_classification/-/raw/master/TP_Classification_v06.zip
Resolving gitlab.in2p3.fr... 134.158.69.41
Connecting to gitlab.in2p3.fr[134.158.69.41]:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 1129440 (1.1M) [application/octet-stream]
Saving to: 'TP_Classification_v06.zip'

TP_Classification_v06.zip      100%[=====]
2026-06-09 18:04:38 (2.02 MB/s) 'TP_Classification_v06.zip' saved [1129440/1129440]

[pmjodoin@nodegpupool1 ~]$ unzip TP_Classification_v06.zip
Archive:  TP_Classification_v06.zip
  inflating: 01_LoadData_K3.ipynb
  inflating: 03_Interpretability_K3.ipynb
  inflating: 04_LearningCurves_K3.ipynb
  inflating: 05_HyperparameterOptimization.ipynb
  inflating: 06_KnowledgeDistillation_K3.ipynb
  inflating: 06_KnowledgeDistillationNoClass_K3.ipynb
  inflating: 09_GradAndAutoGrad.ipynb
  inflating: toolsK3.py
  inflating: figures/architecture_cnn_en.png
  inflating: figures/architecture_fcnn_en.png
  inflating: figures/architecture_resnet_en.png
  inflating: figures/decision_tree.png
  inflating: figures/Figure1_SagittalAxialCoronal_small.png
  inflating: figures/ResNet_TransferLearning.png
  inflating: figures/TensorBoard.png
  inflating: figures/treeL2.png
  inflating: 02a_ClassificationRF_K3.ipynb
  inflating: 02b_ClassificationNN_K3.ipynb
  inflating: 05_HyperparameterOptimization_K3.ipynb
[pmjodoin@nodegpupool1 ~]$
```

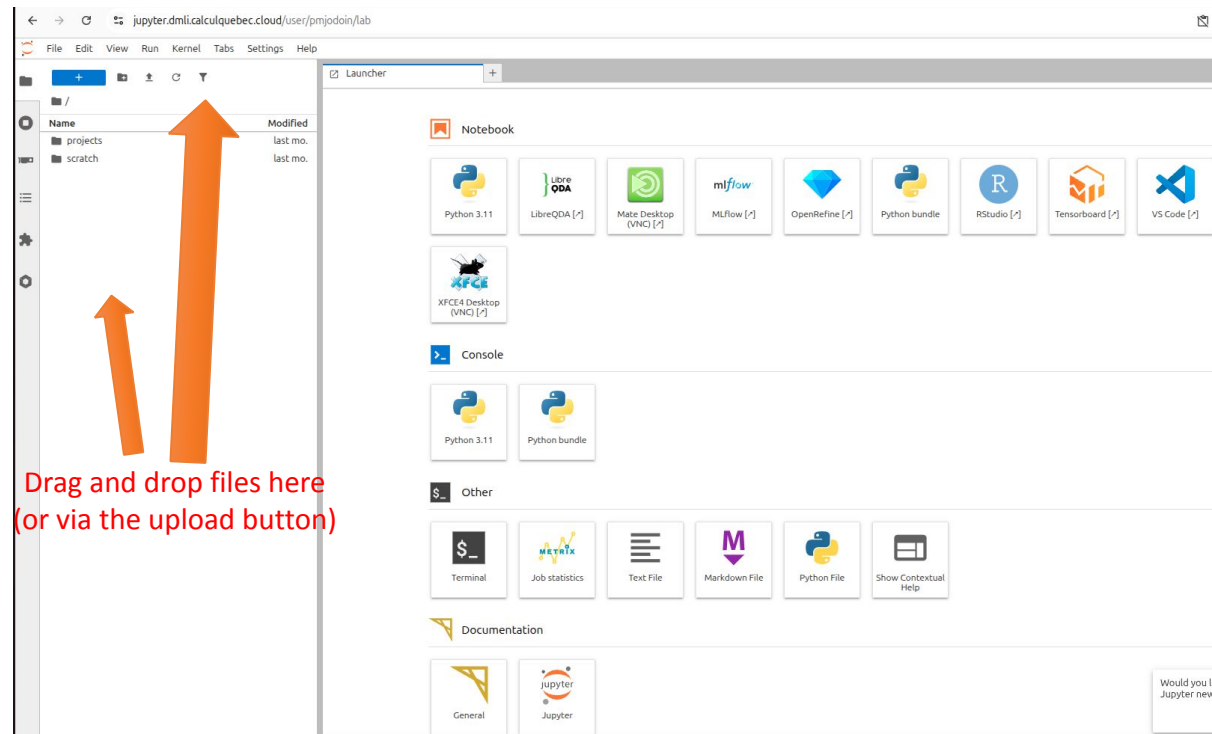
Download Hands-on materials

If the previous step did not work, you can

a) download the zip file on your computer

https://gitlab.in2p3.fr/thomas.grenier/tp1ss_classification/-/raw/master/TP_Classification_v06.zip

a) Upload it in your online environment



Download Hands-on materials

If the previous step did not work, you can

- a) download the zip file on your computer

https://gitlab.in2p3.fr/thomas.grenier/tp1ss_classification/-/raw/master/TP_Classification_v06.zip

- a) Upload it in your online environment

- a) Unzip the file

Hands on 2

For Hands on 2, please focus on the 7 last Notebooks

1. Ha

2. Int

This hands-

And so, this

learning an

MR images

Here, we have to recognize whether a brain slice image is axial, sagittal or coronal and comes from T1w, T2w and Proton Density (PD). So, this is a classification problem with 9 classes, as follows:

Axial Sagittal Coronal

T1